Paper

A systematic review and meta-analysis of the efficacy and safety of pharmacological treatments for obesity in adults. Nature Medicine 2025.

manually curated

Notebook

Background & objective

Obesity is a biologically driven chronic disease with rapidly increasing prevalence. Novel anti-obesity medications (OMMs) are emerging, but direct head-to-head comparisons in randomized controlled trials are still limited. This study aims to use network meta-analysis to evaluate the efficacy and safety of weight-loss drugs.

Tools & dependencie

- pandas
- sklearn
- numpy
- matplotlib
- scipy
- seaborn

Data & descriptions

- Orlistat
- Naltrexone/Bupropion
- Liraglutide
- Phentermine/Topiramate
- Semaglutide
- Tirzepatide

- **weight_loss_tbwl.csv** — Total Body Weight Loss percentage(TBWL)
- **quality_of_life_outcomes.csv** — Patient-reported quality of life
- **metabolic_parameters_blood_pressure.csv** — Metabolic parameters and blood pressure changes
- **mental_health_outcomes.csv** — Mental health adverse events
- **obesity_comorbid_conditions.csv** — Remission of obesity-related comorbidities
- **safety_outcomes.csv** — Overall safety outcomes
- **cardiovascular_effects.csv** — Risk of cardiovascular hard endpoints

Hyposis and Findings

Do these weight-loss drugs show statistically significant effects compared with placebo?

All 6 drugs showed significant weight loss vs. placebo, ranging from 3.1% to 16.2% TBWL.

Can an effectiveness threshold be defined toidentify superior drugs?

Defines three tiers: Tier 1 ≥60 pts, Tier 2 40–60 pts, Tier 3 <40 pts.

What clinical effects and outcomes are associated with each drug?

Tirzepatide & Semaglutide: weight loss + T2D remission + MASH resolution + reduced HFF...